Listen to Detect: Onset-Prompted Personalized Voice Activity Detection

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Abstract. Personalized voice activity detection (PVAD) extends conventional voice activity detection (VAD) from binary speech/non-speech classification to frame-level target-speaker activity detection in multi-speaker mixtures. Existing PVAD systems typically encode enrollment utterance into a compact, fixed-length target-speaker embedding, which is then used to condition frame-level mixture embeddings to realize PVAD. In this context, we propose *Listen to Detect (L2D)*, an onset-prompted PVAD approach which prepends enrollment utterance to mixture and trains a deep neural network (DNN) based PVAD system on the prepended mixture. The prepended enrollment can serve as a variable-length prompt context for the DNN model, potentially resulting in better speaker conditioning than first compressing the enrollment into a fixed-length speaker embedding and then performing speaker conditioning. When VAD labels are available for the enrollment, we further introduce prompt-region supervision, allowing the PVAD model, during training, to be also supervised in the prepended enrollment region. Evaluation results on the noisy SparseLibri2Mix dataset demonstrate that the proposed L2D system achieves state-of-the-art PVAD performance.

Keywords: personalized voice activity detection · target speech processing · speaker conditioning

1 Introduction

Voice activity detection (VAD) is a fundamental front-end component for speech processing tasks such as automatic speech recognition (ASR), speaker recognition, speaker diarization, meeting transcription, and speech enhancement [19, 9, 15, 1, 27, 11]. Personalized VAD (PVAD) extends VAD from detecting whether any speech exists in each frame of a mixture to whether a specific enrolled speaker is active. The output is typically categorized into three classes [7], including target-speaker speech (TSS), non-target-speaker speech (NTSS) and non-speech

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(NS), so PVAD jointly performs speech activity detection and target-speaker discrimination. PVAD becomes especially challenging in mixtures containing multiple speakers and concurrent speech, since speech activity alone does not determine the frame label [13, 8]. A frame should be labeled as target speech only when the enrolled speaker is active, and a frame with speech only from an interfering speaker should be treated as non-target speech. PVAD systems hence need to condition mixture representations on enrollment information. Mainstream systems often perform the conditioning by extracting a compact, fixed-length speaker embedding from an enrollment utterance and injecting it into the detection network [20, 5]. As a result, most PVAD systems follow an enrollment-based conditioning paradigm, where a speaker embedding extracted from the enrollment utterance guides frame-level target-speech activity detection. For example, the Personal VAD system proposed by Ding *et al.* [7] conditions a VAD-like network on a target-speaker embedding or a speaker verification score, and their Personal VAD 2.0 system further develops speaker embedding modulation for streaming on-device ASR [6].

Although speaker conditioning based on fixed-length speaker embedding is effective [14, 7, 6, 21, 4], using the enrollment utterance only through a fixed-length speaker embedding may restrict the information that can be exploited by the PVAD model. The resulting embedding primarily provides a global cue about the identity of the target speaker [12, 18] and may not retain the temporal variations in phonetic and spectral patterns that could help identify when the target speaker is active. To more thoroughly exploit the enrollment utterance, another line of research avoids explicit speaker embeddings and directly models the interaction between enrollment and mixture embeddings, often through attention or cross attention mechanisms [24, 26, 25, 29, 28, 17]. However, these methods usually maintain separate enrollment and mixture branches and rely on explicitly-designed interaction modules.

Motivated by the Listen to Extract (LExt) study [17] designed for target speaker extraction (TSE), which prepends enrollment utterance to mixture and processes the prepended mixture signal with a single DNN model for speaker extraction, we investigate a similar prompting strategy for PVAD. In this formulation, the enrollment utterance, a short separator, and the mixture are concatenated as a single input sequence, allowing the network to use waveform level enrollment context when predicting target speaker activity in the mixture.

The contributions of this work can be summarized as follows:

- We propose an onset-prompted formulation for PVAD, where the enrollment utterance, a separator, and the mixture are concatenated and processed by a single DNN model for PVAD. After the prepending, the target speaker is always the first speaker to speak. This way, the DNN could easily figure out who the target speaker is by checking the speech onset information, and realizes accurate PVAD.
- We propose prompt region supervision for the prompted PVAD setting. When VAD labels are available for the enrollment utterance, the model

can be optimized over all of the enrollment, separator, and mixture regions, rather than only over the mixture region.
– Evaluation results on the noisy SparseLibri2Mix dataset show state-of-the-art performance, supplemented by controls on conditioning, enrollment position and duration, and inference cost.

2 Related Work

2.1 Personalized Voice Activity Detection

In [7], Personal VAD is introduced as a speaker conditioned extension of VAD, aiming to detect the activity of a specified target speaker in applications such as on-device ASR and personalized voice assistants. Instead of detecting arbitrary speech, these systems leverage enrollment speech or a speaker profile, and predict whether the enrolled speaker is active. This target-specification feature is usually implemented by first extracting a fixed-length speaker embedding from the enrollment utterance and then using it to condition a frame-level speaker activity detector. The recent Target-Speaker Speech Processing Universal Performance Benchmark (TS-SUPERB) [14] includes PVAD as one of its target-speech processing tasks and provides official self-supervised learning (SSL) based baselines for standardized evaluation. In summary, existing PVAD systems mainly use enrollment utterance as a speaker identity cue for conditioning a frame-level speaker activity detector, leaving the direct use of enrollment utterance as input context under-explored.

2.2 Enrollment Modeling for Personalized VAD

Existing PVAD systems usually use enrollment speech through a speaker representation that conditions a frame-level activity detector. Recent work has explored alternatives to fixed speaker embeddings. Zeng and Li [28] propose Universal Speaker Embedding Free Target Speaker Extraction and Personal Voice Activity Detection (USEF-TP), extending speaker-embedding-free target speaker modeling to both TSE and PVAD by explicitly modeling the interaction between enrollment and mixture embeddings. Another way exploiting cues in enrollment utterance is to move the enrollment speech directly into the input sequence. Listen to Extract (LExt) is originally proposed for TSE and prepends the enrollment utterance to the mixture with a short separator, allowing a single DNN model to process the concatenated signal [17]. This work adapts the same input prompting principle to PVAD, where the objective is frame-level target speaker activity detection rather than waveform extraction. Waveform concatenation itself is inherited from LExt; our contribution is its evaluation for three-class PVAD and the prompt-region classification objective. Both tasks require enrollment-conditioned speaker discrimination, which motivates using the same input context. However, reconstruction must preserve the target signal, whereas PVAD only needs activity posteriors. The classification loss therefore

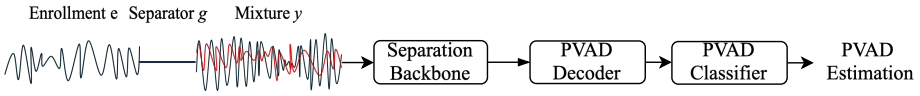


Fig. 1. Illustration of Listen to Detect (L2D). The separator signal is an input waveform segment; the separation backbone is the DNN module. The classifier can be supervised over all three input regions during training, while inference retains only mixture-region predictions.

does not require faithful target waveform reconstruction; prompt-region supervision additionally provides direct TSS/NS training signals on the enrollment. These objective differences motivate the adaptation, but do not establish that onset prompting is optimal or that the learned representations match those of TSE.

3 Method

3.1 Problem Formulation

Let $y \in \mathbb{R}^N$ denote a speech mixture signal with N samples and $e \in \mathbb{R}^{N_e}$ an enrollment utterance (with N_e samples) from the target speaker. The mixture may contain speech from the target speaker, speech from one or more non-target speakers, and background noises. The enrollment utterance specifies the speaker of interest, and the goal of PVAD is to determine whether this speaker is active in each frame of the mixture.

We denote the frame level PVAD label sequence of the mixture as $\mathbf{z} = \{z_t\}_{t=1}^{T}$, where T is the number of frames in mixture. Each label belongs to one of three classes:

$$z_t \in \{\mathrm{TSS}, \mathrm{NTSS}, \mathrm{NS}\},$$

where TSS denotes target speaker speech, NTSS non-target speaker speech, and NS non speech. A frame is labeled as TSS whenever the enrolled speaker is active, even if other speakers are also active. It is labeled as NTSS when speech is present but the enrolled speaker is inactive, and as NS when no speech is present.

Given the mixture y and enrollment utterance e, a PVAD system estimates the posterior probability of the three classes for each mixture frame:

$$\mathbf{p}_t = [P(z_t = k \mid y, e)]_{k \in \mathcal{C}}, \quad t = 1, \ldots, T. \tag{1}$$

The predicted label is obtained by selecting the class with the highest posterior probability:

$$\hat{z}_t = \underset{k \in \{\mathrm{TSS},\mathrm{NTSS},\mathrm{NS}\}}{\arg\max} \; p_{t,k}. \tag{2}$$

3.2 Onset-Prompted PVAD

We propose an onset prompted formulation for PVAD, in which the enrollment utterance is prepended to the mixture at the waveform level to form a single input sequence. Specifically, a short separator s is inserted between the enrollment utterance e and mixture y. The prompted waveform is constructed as

$$\widetilde{y} = [e; s; y], \tag{3}$$

where $[\cdot;\cdot]$ denotes concatenation along the time dimension. Here $s = g\mathbf{1}_{N_s}$ is a constant-valued, non-speech separator signal: g is its sample amplitude and N_s its sample count, set to 320 (20 ms at 16 kHz) by default. Thus, g denotes the amplitude of the signal s in Eq. 3, not another network module. Setting $g = 0$ retains a silent gap; only $N_s = 0$ removes the separator signal completely. We use "separation backbone" for the DNN module to distinguish it from this waveform segment. The separator marks the boundary between the enrollment and mixture regions and provides an explicit indication of transition between the two signals. An overview of the proposed onset prompted PVAD framework is shown in Fig. 1. The resulting sequence is processed by a single backbone, allowing the model to use the enrollment waveform directly as context when predicting target speaker activity in the mixture region.

Unlike speaker embedding conditioning, this formulation preserves the temporal structure of the enrollment utterance and allows the backbone to exploit its phonetic and spectral variations when detecting target speaker activity. For subsequent description, we divide the prompted sequence into three regions, denoted by $\mathcal{T}_e$, $\mathcal{T}_s$, and $\mathcal{T}_y$, corresponding to the enrollment utterance, separator, and mixture, respectively. The model produces frame-level predictions over the complete prompted sequence, while only the predictions indexed by $\mathcal{T}_y$ are retained during PVAD inference.

3.3 DNN Backbone

To evaluate the proposed onset prompting formulation with different backbone DNN models, we employ TF-GridNet [22, 23] and TF-Locoformer [16] for the proposed onset prompting framework. TF-GridNet is originally proposed for speech separation. It models spectral and temporal dependencies within a unified network. TF-Locoformer adopts a Transformer architecture that combines convolution with self-attention to capture local patterns and global context without recurrent layers. For PVAD, we use their target-related spectral outputs as features for a classification head, which predicts the probabilities of TSS, NTSS and NS at each frame. For TF-GridNet, the head concatenates the log-magnitude input and estimated spectra, followed by projection, normalization, PReLU, a single-layer BLSTM, and a three-class linear output. Training uses frame-level cross entropy rather than a waveform reconstruction loss.

3.4 Prompt Region Supervision

The onset prompted model produces frame-level predictions for the enrollment, separator, and mixture regions. When enrollment VAD labels are available, we apply supervision to all three regions during training. For the enrollment utterance, speech frames are labeled as TSS and non-speech frames are labeled as NS. Since the separator does not contain any speech, all separator frames are labeled as NS. The mixture region retains its original TSS, NTSS, and NS labels. During inference, only the predictions in the mixture region are used as the final output. In our experiments, enrollment VAD labels are automatically generated pyannote VAD pseudo-labels, rather than manual frame-level annotations. If these labels are unavailable, the enrollment remains input context and the loss is restricted to the mixture region, as in Table 3. Neither variant needs enrollment VAD labels at inference. Pseudo-label quality is not separately evaluated here.

4 Experimental Setup

4.1 Dataset

We leverage the PVAD setup in TS-SUPERB [14], which is based on the SparseLibriMix dataset [3], to evaluate the effectiveness of L2D. We follow its official training, validation, test, and enrollment protocols. The test set comprises four conditions with 0%, 20%, 40%, and 60% speaker overlap, each consisting of $1,000$ enrollment-conditioned mixtures. Each mixture consists of one enrolled target speaker speech and one non-target speaker speech. Depending on the overlap condition, zero, one, or both speakers may be active in each frame. For every mixture frame, our model predicts one of the three labels: TSS,NTSS, and NS.

4.2 Model Configurations and Baseline Systems

We investigate two DNN architectures, TF-GridNet [23, 22] and TF-Locoformer [16], to build our L2D model. Both architectures obtain strong performance in recent speech enhancement and separation benchmarks, and are adopted by many studies. This study extends them for PVAD task. We compare L2D performance with USEF-TP [28], a recent PVAD system, and the TS-SUPERB results [14], which are based on SSL embeddings. This section describes each of the systems in detail.

L2D with TF-GridNet. The L2D TF-GridNet configurations follow the LExt study [17]. We consider two TF-GridNet configurations, denoted respectively as TF-GridNetV1 and TF-GridNetV2. In the V1 setting, we configure the number of TF-GridNet blocks to 4, the embedding dimension to 128, and the number of BLSTM hidden units to 200. In the V2 setting, we increase the number of TF-GridNet blocks to 6, and the number of BLSTM hidden units to 256, with the remaining settings same as V1. In L2D, the original mixture waveform and the

enrollment utterance are independently RMS-normalized before prepending. We denote this pre-processing technique as root-mean-square (RMS) normalization, and is performed if not specified. The separator signal is set to 20 ms, and is filled with a constant value $g = 5$.

L2D with TF-Locoformer. For the L2D TF-Locoformer configuration, we use the medium setup from the original TF-Locoformer paper [16]. That is, 6 Locoformer blocks, a 128-dimensional embedding, a 384-dimensional Conv-SwiGLU hidden layer, 4 self-attention heads, 4 RMSGroupNorm groups, and a convolution kernel size of 4. Note that only the DNN backbone is changed. The rest settings, including onset prompting, RMS normalization, separator construction, and the PVAD head, remain the same as the L2D TF-GridNet system described in the previous subsection.

USEF-TP. Our USEF-TP implementation follows the network configuration of Zeng and Li [28]. The cross multi-head attention module has one attention layer with 4 heads and a 512-dimensional feed-forward layer. Its separation backbone contains 6 TF-GridNet blocks with 256 BLSTM hidden units. We retain the enrollment–mixture cross-attention used for speaker conditioning and replace the original joint TSE/PVAD decoders with the three-class PVAD head described in the original USEF-TP [28].

TS-SUPERB PVAD System. The PVAD system in TS-SUPERB [14] is trained based on the SSL embeddings extracted from the input mixture by HuBERT [10] or WavLM [2]. The motivation is that modern SSL models, through large-scale pre-training on massive single- or multi-speaker mixture signals, can provide discriminative embeddings that could benefit downstream target-speech processing tasks. The PVAD system in TS-SUPERB attaches a light-weighted BLSTM based classification head to the SSL embeddings, and is trained to perform PVAD using the SSL embeddings and a speaker embedding extracted from enrollment utterance.

4.3 Training Details

All the L2D and USEF-TP models are trained for $60,000$ steps with Adam, using a learning rate starting at 10^{-4}, a gradient clipping value set to 1, and a batch size of 8. At 16 kHz sampling rate, we use a window size of $1,024$ samples, a hop size of 256 samples, and a square-root Hann analysis window. The best checkpoint is selected according to the performance on the official validation set. During training, the mixture signals used for training are limited to 192 label frames, as the memory of our GPUs is limited. Final evaluation covers the complete test utterance, first attempting full-utterance inference and falling back to 192-frame blocks if GPU memory is insufficient. The four re-evaluated

TF-GridNet/USEF-TP systems and completed conditioning controls used full-utterance inference for all 4,000 enrollment-conditioned test inputs per system, with no fallback.

For the PVAD system in TS-SUPERB, we directly use its pre-trained model for comparison, as we strictly follow its training and test configurations.

4.4 Evaluation Metrics

We evaluate every test utterance over its full duration, and report averaged scores over the four overlap conditions. Following the TS-SUPERB protocol [14], we report mean average precision (mAP), defined as the arithmetic mean of average precision (AP) over the classes: TSS, NTSS, and NS. We additionally report frame-level accuracy and target-active F1, precision, and recall scores, where TSS is treated as the positive class and NTSS/NS as the negative class. Accuracy is computed from the maximum-posterior class, whereas AP-based metrics use posterior probabilities.

5 Results and Discussions

5.1 Main Results and Comparison with Baseline Systems

Table 1 compares the proposed L2D system with official TS-SUPERB PVAD results. L2D with TF-GridNetV2 obtains the best result, reaching 98.03% macro mAP. It outperforms TS-SUPERB PVAD systems which use HuBERT Large and WavLM Large embeddings, as well as USEF-TP, a strong PVAD model also based on TF-GridNet. These results show that both explicit enrollment–mixture attention and waveform-level onset prompting are effective. The V2 advantage over USEF-TP is 0.26 percentage points of mAP; this single-checkpoint comparison does not establish statistical significance or isolate conditioning from architecture differences. With TF-GridNetV2, L2D also outperforms L2D with TF-GridNetV1. This is reasonable as TF-GridNetV2 uses more computation. The retrained TF-Locoformer (M) obtains 97.17% average mAP under the same evaluation protocol. The official SSL-BLSTM baselines and L2D differ in pre-training, backbone, and parameter budget. Table 1 is therefore a benchmark comparison rather than a capacity-matched test of enrollment conditioning. The same-backbone fixed-vector control below addresses one part of this issue, but does not substitute for an SSL-conditioned TF-GridNet comparison.

Table 1. PVAD results on SparseLibri2Mix. mTSS denotes mean target-speech AP. F1, precision, and recall scores use target speech as the positive class. All scores are reported in percentage (%).

Model	Backbone	#Params (M)	mAP	mTSS	Acc.	F1	Prec.	Rec.
TS-SUPERB HuBERT Large [14]	SSL-BLSTM	–	95.4	97.3	–	–	–	–
TS-SUPERB WavLM Large [14]	SSL-BLSTM	–	96.6	98.0	–	–	–	–
USEF-TP [28]	TF-GridNet	16.37	97.77	98.88	92.71	94.24	92.29	**96.33**
L2D	TF-GridNet (V1)	6.13	97.99	99.01	93.62	94.89	94.28	95.51
L2D	TF-GridNet (V2)	12.40	**98.03**	**99.03**	**93.66**	**95.01**	**94.76**	95.30
L2D	TF-Locoformer (M)	15.20	97.17	98.18	92.10	93.55	91.55	95.68

5.2 Ablation Results

Table 2 reports the results of using RMS normalization and setting the separator value g to various values, based on L2D with TFGridNetV1. Table 3 compares mixture-only and mix+enrollment supervision. All variants have the same number of trainable parameters. This section describes them in detail.

RMS Normalization and Separator Value. Before concatenation, we optionally apply root-mean-square (RMS) normalization to each valid waveform independently. Specifically, for a waveform u, we use $\tilde{u} = u/(\sqrt{\frac{1}{L}\sum_{l=1}^{L} u_l^2} + \epsilon)$, where L is the valid waveform length and ϵ prevents numerical instability. The separator signal s is a short constant-valued waveform inserted between the enrollment and mixture signals; its amplitude is controlled by g.

Table 2 shows that RMS normalization is the main contributor to robust performance. At $g = 5$, enabling RMS normalization improves mAP from 94.49% to 97.13% and F1 from 90.99% to 94.42%. Among the reported configurations, RMS with $g = 0$ achieves the best overall mAP (97.55%) and F1 (94.43%). In contrast, increasing the separator value from 0 to 5 under RMS normalization raises precision from 92.84% to 93.79%, but reduces recall from 96.10% to 95.09%. Therefore, the separator value mainly controls the precision–recall trade-off, whereas RMS normalization provides the larger improvement in these configurations. These amplitude ablations alone do not isolate the contribution of waveform prompting, and a nonzero separator amplitude does not consistently improve mAP.

Table 2. Effects of using RMS normalization and separator values on L2D with TF-GridNetV1. Scores are reported in percentage (%). "–" denotes unavailable results.

System	RMS	Separator Value g	mAP	mTSS	Acc.	F1	Prec.	Rec.
L2D	✓	0.0	**97.55**	**98.71**	93.00	**94.43**	92.84	**96.10**
L2D	×	5.0	94.49	97.16	88.59	90.99	88.34	93.87
L2D	✓	5.0	97.13	98.68	**93.08**	94.42	93.79	95.09
L2D	×	0.0	96.56	98.52	92.50	94.00	**94.46**	93.58

Mix-Only vs. Mix+Enrollment Supervision. The prompt-region experiment extends the original L2D objective by assigning frame-level labels to the enrollment region instead of using it only as prompt context without any labels. As shown in Table 3, this change improves mAP from 97.13% to 97.99% and F1 from 94.42% to 94.89%. It also improves precision from 93.79% to 94.28% and recall from 95.09% to 95.51%. This result indicates that the artificial onset is useful not only as conditioning context but also as a directly supervised example of the target speaker.

Table 3. Supervision-region ablation of L2D with TFGridNetV1 (using RMS normalization and with separator value g set to 5). Scores are reported in percentage.

Loss region	mAP	mTSS	Acc.	F1	Prec.	Rec.
Mixture only	97.13	98.68	93.08	94.42	93.79	95.09
Mixture + enrollment	**97.99**	**99.01**	**93.62**	**94.89**	**94.28**	**95.51**

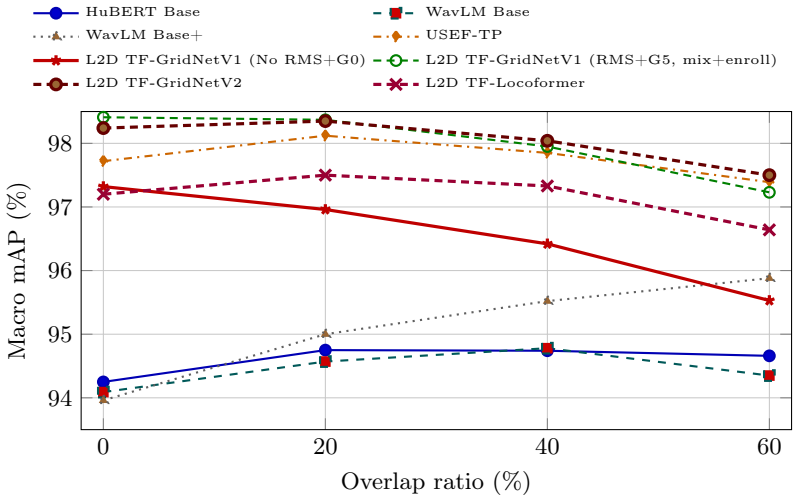


Fig. 2. Macro mAP vs. speaker overlap ratios.

5.3 Analysis by Overlap Ratio and Class

Fig. 2 shows the mAP trend as the speaker overlap ratio increases, and Table 4 consolidates the corresponding class-wise results for all completed systems. All L2D variants maintain high TSS AP across the four test conditions. The main degradation under heavier overlap occurs in NTSS AP, indicating that rejecting an active interfering speaker is the most difficult part of PVAD.

Table 4. Class-wise AP (%), mAP (%), and accuracy (%) in various speaker overlap conditions.

Model	Overlap	TSS AP	NTSS AP	NS AP	mAP	Acc.
L2D TF-GridNetV1 (No RMS+G0)	0%	98.61	98.44	94.92	97.32	94.10
	20%	98.31	96.25	96.31	96.96	92.68
	40%	98.52	94.56	96.18	96.42	92.06
	60%	98.63	91.89	96.06	95.53	91.18
	Avg.	98.52	95.28	95.87	96.56	92.50
L2D TF-GridNetV1 (RMS+G5, mix+enroll)	0%	99.11	98.91	97.22	98.41	94.46
	20%	98.94	97.54	98.63	98.37	93.62
	40%	98.99	96.15	98.72	97.95	93.43
	60%	98.99	93.82	98.88	97.23	92.98
	Avg.	99.01	96.60	**98.36**	97.99	93.62
L2D TF-GridNetV2	0%	98.93	98.68	97.09	98.24	94.13
	20%	98.92	97.53	98.61	98.35	94.05
	40%	99.12	96.55	98.45	98.04	93.39
	60%	99.16	94.80	98.53	97.50	93.07
	Avg.	**99.03**	**96.89**	98.17	**98.03**	**93.66**
USEF-TP [28]	0%	98.52	98.26	96.38	97.72	92.03
	20%	98.86	97.36	98.14	98.12	92.81
	40%	99.00	96.27	98.27	97.85	93.00
	60%	99.16	94.59	98.41	97.39	93.02
	Avg.	98.88	96.62	97.80	97.77	92.71
L2D TF-Locoformer (M)	0%	97.31	97.23	97.07	97.20	91.39
	20%	97.96	95.83	98.72	97.50	91.95
	40%	98.68	94.56	98.76	97.33	92.64
	60%	98.78	92.28	98.87	96.64	92.41
	Avg.	98.18	94.97	**98.36**	97.17	92.10

RMS+G5 with mix+enrollment supervision raises the average NS AP of TF-GridNetV1 from 95.87% to 98.36% and its mAP from 96.56% to 97.99%. Switching the backbone from TFGridNetV1 to TF-GridNetV2 provides a further gain: it reaches 98.03% average mAP and gives the strongest average TSS and NTSS AP among the completed systems. USEF-TP remains competitive across all speaker overlap conditions, confirming that explicit enrollment–mixture attention is also effective.

5.4 Enrollment Controls

Conditioning and Prompt Position. Table 5 compares newly trained controls with the same TF-GridNetV1 backbone, STFT, PVAD head, RMS normalization, mixture-only loss, optimizer, training data, seed 1337, 60,000 updates, and best-dev checkpoint selection. These controls are distinct from the original checkpoints in Tables 1 and 3. The fixed-vector control averages enrollment log-magnitude STFT features over valid frames, maps them to a 128-dimensional vector with an MLP, and uses vector-derived scale and bias to condition the mixture STFT before TF-GridNet. Its enrollment encoder is jointly trained from scratch, without speaker-verification pretraining or a speaker-ID loss; it adds 0.43M parameters and is not an exact parameter match.

The waveform-prepending control achieves 97.60% mAP versus 95.18% for the fixed vector at seed 1337, supporting direct waveform context under this controlled setting. This result does not establish superiority over pretrained speaker

Table 5. Conditioning and position controls with TF-GridNetV1, input RMS normalization, mixture-only supervision, seed 1337, and 60,000 training steps. Scores are percentages; – denotes a result awaiting complete evaluation.

Conditioning	Params (M)	mAP	TSS AP	Acc.	F1	Prec.	Rec.
Prepend, $g = 5$, 20 ms	6.13	97.60	98.78	92.51	93.94	92.25	95.72
Prepend, no separator	6.13	97.30	98.80	93.02	94.40	**94.01**	94.81
Append, $g = 5$, 20 ms	6.13	**98.04**	**98.96**	**93.50**	**94.73**	93.39	**96.13**
Fixed 128-D vector	6.56	95.18	96.68	88.91	90.65	90.77	90.56

embeddings. The control evaluates activity detection, without an empirical comparison of the learned TSE/PVAD representations. Removing the separator signal entirely yields 97.30%, while appending enrollment yields 98.04%. Thus, neither a nonzero boundary signal nor the onset position is required for competitive offline PVAD. The appended model can use later enrollment context through the bidirectional backbone, which may help explain its slightly higher score. Evaluation uses one utterance per batch, so padding between utterances cannot account for the difference. Nevertheless, changing the order also changes which mixture boundary is exposed to STFT and convolutional padding, as well as the alignment of mixture frames. These effects have not been isolated, and the 0.44-point advantage comes from a single training seed; we therefore do not attribute it to enrollment order alone. This comparison does not demonstrate causal or streaming operation.

Table 6. Average mAP (%) across four overlap conditions for seeds 1337, 2027, and 2028. Mean and sample standard deviation are reported only after all three seeds complete; – denotes pending results.

Conditioning	1337	2027	2028	Complete	Mean ± SD
Prepend waveform	97.60	97.39	98.06	3/3	97.68 ± 0.34
Fixed 128-D vector	95.18	94.65	95.05	3/3	94.96 ± 0.28

Table 6 reports three independent training seeds for each conditioning method; the standard deviation describes training variability and does not establish statistical significance against other systems.

The fixed-vector control obtains 94.96 ± 0.28% mAP over three seeds (sample standard deviation). The waveform-prepending control obtains 97.68 ± 0.34% mAP over the same three seeds (sample standard deviation). The conditioning-control seed coverage is reported in Table 6; the other model comparisons remain single-seed evaluations, and small differences should be interpreted descriptively.

Enrollment Duration and Invalid Enrollment. Table 7 evaluates waveform prefixes of 1, 2, and 4 seconds using the original TF-GridNetV1 RMS+$g = 5$

mix+enrollment checkpoint, without retraining. mAP increases from 89.93% at 1 second to 97.42% at 4 seconds, approaching 97.99% with full enrollment. Prefix cropping changes phonetic content and silence as well as duration, so this is not a controlled comparison of voiced duration alone.

Table 7. Enrollment-prefix duration using the same TF-GridNetV1 RMS+$g = 5$ mix+enrollment checkpoint. Average mAP is over four overlap conditions. Prefix duration does not measure voiced duration.

Enrollment	mAP (%)
1 s	89.93
2 s	94.08
4 s	97.42
Full enrollment	97.99

We also replace enrollment with speech from a third speaker absent from the mixture, or a one-second zero waveform as a proxy for silent or missing registration. In these conditions no valid TSS frames exist. We therefore report the false-target frame rate, defined as the fraction of all valid mixture frames whose maximum-posterior prediction is TSS, rather than target AP or mAP.

Table 8. Invalid-enrollment controls using the original TF-GridNetV1 mix+enrollment checkpoint. False-target frame rates (%) are over all valid mixture frames; Avg. is the arithmetic mean over overlap conditions.

Enrollment	0%	20%	40%	60%	Avg.
Absent third speaker	35.72	46.38	48.76	50.39	45.31
One-second zero waveform	85.85	88.27	87.68	87.49	87.32

Table 8 shows substantial false target activity under both controls. The evaluated model therefore depends on valid enrollment and does not reliably reject absent targets or abstain under silent registration. These tests identify failure modes; they do not demonstrate robustness to invalid enrollment.

5.5 Inference Cost

We measure inference on one RTX 5880 Ada GPU with batch size one, using 10 warm-up inputs and the next 100 inputs from the 0%-overlap split. Let t_i be the model-call time in seconds for input i, and d_i the duration in seconds of its mixture waveform. For $K = 100$ timed inputs, the real-time factor (RTF) and

Table 9. Offline inference on one RTX 5880 Ada GPU: batch size one, 10 warm-up and 100 timed utterances, with CUDA synchronization. Memory is peak CUDA allocated memory in GiB; call time includes input/output transfers and is not streaming latency.

Model	Params (M)	RTF	Peak GiB	Mean call (ms)
TF-GridNetV1 (mix+enroll)	6.13	0.0492	13.28	444.6
TF-GridNetV2	12.40	0.0806	16.29	728.0
USEF-TP	16.37	0.0173	3.64	155.8

mean model-call time are

$$\mathrm{RTF} = \frac{\sum_{i=1}^{K} t_i}{\sum_{i=1}^{K} d_i}, \qquad \bar{t}_{\mathrm{call}} = \frac{1000}{K} \sum_{i=1}^{K} t_i \quad \text{(ms)}. \tag{4}$$

An RTF below one means that the measured processing time is shorter than the mixture duration. Timing uses CUDA synchronization and includes prompt processing and input/output transfers, excluding checkpoint and data loading. Table 9 reports RTF and mean call time alongside model parameter count and peak CUDA allocated memory to characterize runtime and memory cost. All three systems have RTF below one, although V1 is slower and uses more memory than USEF-TP despite having fewer parameters; V2 further increases runtime and memory. These are offline measurements and do not establish streaming latency.

6 Conclusion

We have proposed L2D, an onset-prompted PVAD approach. L2D adapts waveform-level onset prompting from LExt to three-class PVAD and adds optional prompt-region supervision. It achieves state-of-the-art results with USEF-TP and the official TS-SUPERB PVAD baselines on SparseLibri2Mix. Same-backbone controls support using waveform context over the jointly trained fixed-vector encoder tested here, while position and invalid-enrollment controls clarify the limits of the onset formulation. These findings suggest that target-speaker conditioning style is a key design choice when adapting DNN models designed for speech separation to PVAD.

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